IMPLEMENTATION HANDOVER / AGENT INPUT

PUKfm Radio CRM

Complete SuiteCRM 8 Rebuild, Quote Scheduling, Campaign Conversion & Myriad Export Specification

Primary purpose: Provide a self-hosted AI engineering agent with the exact known design, prior file locations, business rules, installation order, verification requirements and recovery history needed to reproduce the Radio CRM workflow.

Project version: Technical handover reconstructed from the project conversation and previously reported working tests. It is NOT a verified copy of the original source repository.

Last known application: SuiteCRM 8.10.1 / Ubuntu 24.04 LTS / PHP 8.3 / MariaDB 10.11 / Apache. Last supplied public IP: 92.4.131.128 (verify at deployment).

Prepared: 16 October 2026. Audience: local AI agent, developer and system administrator.

RECONSTRUCTION RULE

Recover and inspect any surviving original files or verified backups first. Use this PDF as the behavioural and architectural specification. Do not claim a script is identical to the previously working version without a source diff and passing end-to-end tests.

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Each section names the expected feature, known implementation path, required behaviour, and how to verify it. The appendix contains a master execution prompt that can be supplied directly to an autonomous local agent.

1. Executive specification and evidence levels

Build a radio-advertising workflow for PUKfm inside SuiteCRM 8. Users create a Quote, select a campaign date range and recurring weekly advert slots, receive calculated product line items and a clean Quote PDF, then use the existing Quote action to convert or update the linked Radio Campaign. The campaign stores cart number and exports schedule-accurate CSV and daily Myriad traffic log files.

Component	Previously confirmed	Fresh-install status / required action
SuiteCRM 8 base installation	8.10.1 installer reported completion; local HTTP/1.1 200 OK	Confirm external access, login, Quotes and database health
Quote weekly schedule	Worked on old instance with 24 hourly rows and date-aware totals	Fresh script wrote only a basic template; integration and date totals must be rebuilt
Automatic Quote products	Working examples of priced product rows and VAT	Fresh script created catalog items but omitted production hook/AJAX sync
Quote to Campaign	Existing action and linked Campaign creation reported working	Fresh script may create duplicates; restore idempotent behaviour
SuiteCRM 8 Myriad action	Working Actions-menu export on old instance	Fresh script used a simplified handler that may redirect/exit inside process
Exports	Verified scheduled-play CSV and daily .log files on old instance	Repeat tests and inspect file content, folder permissions
Old data / backups	Backup workflow discussed and script previously created	Not confirmed present on new instance; do not assume old data exists

> Never equate file creation with feature completion
>
> A successful PHP syntax check or Quick Repair is only a preliminary check. Each critical workflow must be tested through the SuiteCRM UI and its persisted database/export effects.

2. Server, installation prerequisites and access

2.1 Known environment

Property	Expected value / requirement
Operating system	Ubuntu 24.04.4 LTS on Oracle Cloud instance (reported)
Application version	SuiteCRM 8.10.1 in reported fresh install; verify installed version
Application root	/var/www/html
Legacy root	/var/www/html/public/legacy
Database	suitecrm_db; prior installer created local suitecrm_user account
Web user / group	www-data / www-data
PHP	8.3 observed in installation log; determine active Apache/FPM integration
Database engine	MariaDB 10.11 observed
HTTP URL	http://92.4.131.128 (last supplied public IP; not independently validated)
Myriad export root	/srv/myriad_exports and /srv/myriad_exports/myriad_daily_logs

2.2 Preflight discovery (do not modify anything yet)

```
sudo bash -c 'date; hostname -I; php -v | head -2; mysql --version'
sudo systemctl is-active apache2 mariadb
sudo apachectl -S
sudo ss -ltnp | grep -E ':80|:443' || true
cd /var/www/html && sudo -u www-data php bin/console --version
sudo mysql -e "SHOW DATABASES LIKE 'suitecrm_db';"
sudo find /srv/radio_crm_backups -maxdepth 3 -type f 2>/dev/null | head
sudo ls -la /usr/local/sbin/restore_radio_crm_custom.sh 2>/dev/null
```

Check Apache DocumentRoot points to /var/www/html/public, rewrite is enabled, the server listens on port 80, and external connectivity is permitted in BOTH host firewall and Oracle VCN Security List/NSG. A local curl -I http://127.0.0.1 result of 200 is not proof that the public IP is reachable.

2.3 Installation order

1. Install and verify the base SuiteCRM 8 release before adding any Radio CRM customizations. Sign in as an admin and confirm the standard Quotes and Product Catalogue features load.
2. Back up the fresh application and database, record SuiteCRM/PHP/DB versions, and establish a rollback checkpoint.
3. Discover surviving original customizations/backup archives before writing replacement files.
4. Register the Radio Campaign module and language, then install database schema/extensions using safe, repeatable migrations.
5. Add Quote date and schedule fields; register Quote UI; build calculations and product sync; add conversion and export service.
6. Run Quick Repair and Rebuild; execute generated SQL only after inspection; clear SuiteCRM 8 and legacy caches; run the acceptance suite.

2.4 Security constraints

- Do not embed static administrator or database passwords from past chat commands in agent output, scripts, a repository, or this document. They were disclosed and must be rotated.
- Prefer HTTPS via a hostname and valid TLS certificate before production usage; restrict SSH, DB and Samba network access.
- Never blindly run DROP DATABASE, rm -rf against non-cache application directories, or a broad recursive chmod that may corrupt permissions.
- Use sudo/service execution only for necessary operations; let www-data own application-managed cache and exports.

3. Required application architecture

End-to-end flow: Quote EditView -> weekly schedule inputs and campaign dates -> accurate total spots -> Quote product lines and amounts -> Quote Save/PDF -> existing Convert to Radio Campaign action -> linked Radio Campaign -> Myriad cart value -> SuiteCRM 8 Export to Myriad record action -> detailed campaign CSV plus daily log files.

3.1 Core module and files

Path relative to /var/www/html	Purpose
public/legacy/modules/RT_RadioCampaigns/	Module class, vardefs, language, metadata and Menu.php
public/legacy/custom/modules/AOS_Quotes/	Quote EditView/DetailView overrides and schedule template
public/legacy/custom/Extension/modules/AOS_Quotes/Ext/	Additional vardefs and labels for date/schedule fields
public/legacy/custom/modules/AOS_Quotes/RadioQuoteScheduleProducts.php	Schedule -> product-line logic hook implementation
public/legacy/custom/modules/AOS_Quotes/Ext/LogicHooks/logichooks.ext.php	Generated hook registry (source should be in Extension/...)
public/legacy/custom/radio_schedule_ajax_sync.php	Live Quote product sync AJAX endpoint
public/legacy/custom/convert_quote_to_radio_campaign.php	Existing Quote-to-Campaign conversion entry point
public/legacy/custom/export_radio_campaign_to_myriad.php	Legacy export reference; only when intentionally used
extensions/defaultExt/config/modules/RT_RadioCampaigns/recordview/actions/export-to-myriad.php	SuiteCRM 8 record action registration
extensions/defaultExt/modules/RT_RadioCampaigns/Process/Service/RecordActions/ExportToMyriadAction.php	SuiteCRM 8 process handler that executes export

Generated aggregate extension files under Ext/ may be overwritten during rebuild. Store authored source in Extension/ paths, rebuild, then inspect the generated results.

4. Radio Campaign module: data model and registration

4.1 Module identity

Module name/key: RT_RadioCampaigns. Bean: RT_RadioCampaigns. Main table: rt_radiocampaigns. Display label: Radio Campaigns. Register in $beanList, $beanFiles, $moduleList and $app_list_strings['moduleList']. Extend SuiteCRM-generated bean/vardef conventions appropriate to SuiteCRM 8.10.1; do not invent a module structure incompatible with the runtime.

4.2 Primary fields

Column / field	Type	Purpose / default
id	CHAR(36) PRIMARY KEY	Bean UUID
name	VARCHAR(255)	Radio Campaign - [Quote Name]
start_date; end_date	DATE	Dates copied from linked Quote
number_of_spots	INT	Total real scheduled plays across date range
spot_length	VARCHAR/enum	30 sec default; options 15, 30, 45, 60 sec
status	VARCHAR/enum	Draft / ready_to_export / exported_to_myriad
myriad_cart_number	VARCHAR(255)	Required before export
quote_reference	VARCHAR(255)	Visible Quote number/reference
linked_quote_id	CHAR(36)	Source Quote ID; needed for schedule lookup
description; assigned_user_id	Standard fields	Notes and ownership
deleted, date_entered, date_modified, etc.	Standard bean fields	SuiteCRM-compatible audit/ownership metadata

Also support rt_radiocampaigns_cstm if SuiteCRM custom-fields infrastructure requires it. A relationship join table previously used was rt_radiocampaigns_aos_quotes_c with join columns rt_radiocampaigns_aos_quotesrt_radiocampaigns_ida and rt_radiocampaigns_aos_quotesaos_quotes_idb. Check actual relationship metadata before defining it.

4.3 Module screens

- ListView columns: name (clickable), start date, end date, number of spots, status, Myriad cart number.
- DetailView and EditView: name, owner, dates, total spots, spot duration, status, cart number, Quote reference/link, description.
- Menu: Create Radio Campaign and View Radio Campaigns; visible in More and relevant module navigation.
- Export action must appear in the actual SuiteCRM 8 DetailView Actions dropdown, not as a blank legacy body control.

5. Weekly Quote schedule: definitive business layout

The weekly advert grid belongs on AOS_Quotes EditView, not on Radio Campaign EditView. It has eight radio programming slots and 24 one-hour rows. Each row has seven numeric input fields for Mon to Sun. This is 24 x 7 = 168 schedule fields. Accept nonnegative integer spot counts; blank fields count as zero.

Slot	Hour field suffixes	Catalogue code / rate
Pre Breakfast	0500_0600	PRE-BREAKFAST / R300
Morning Buzz	0600_0700; 0700_0800; 0800_0900	MORNING-BUZZ / R320
Midmorning Banter	0900_1000; 1000_1100; 1100_1200	MIDMORNING-BANTER / R350
Midday Pulse	1200_1300; 1300_1400; 1400_1500	MIDDAY-PULSE / R350
Cruise Control	1500_1600; 1600_1700; 1700_1800	CRUISE-CONTROL / R350
Specialty Show 1	1800_1900; 1900_2000; 2000_2100	SPECIALTY-SHOW-1 / R300
Specialty Show 2	2100_2200; 2200_2300; 2300_0000	SPECIALTY-SHOW-2 / R220
Music Unlimited	0000_0100; 0100_0200; 0200_0300; 0300_0400; 0400_0500	MUSIC-UNLIMITED / R120

5.1 Exact field naming

The full custom database field naming convention is {day}_{slot}_{hour}_c; day keys are mon tue wed thu fri sat sun. Example actual column names:

```
mon_morning_buzz_0600_0700_c
wed_music_unlimited_0200_0300_c
sun_specialty_show_2_2300_0000_c
```

There must be exactly 168 unique schedule fields on aos_quotes_cstm, plus two date columns campaign_start_date_c and campaign_end_date_c. Use schema inspection to avoid duplication; generated vardefs must match the columns. On a clean SuiteCRM install the aos_quotes_cstm table may not yet exist; create it through supported custom field metadata/repair or a safe migration with primary key id_c.

5.2 Layout and appearance

- At the top of the Quote schedule area, render campaign date fields, then summary: Total Spots: X and Weekly spots: Y.
- The table header is Slot, Time, Mon, Tue, Wed, Thu, Fri, Sat, Sun; group hour rows under each slot.
- Wrapper ID: weekly_advert_schedule_wrapper. Summary IDs: pukfm_weekly_total_bar, pukfm_weekly_total, radio_total_spots_detail.
- Each numeric input uses its exact field name as HTML name and ID. Preserve saved values when re-opening an existing Quote.
- Clear All Spots must set all schedule inputs to zero, update totals, trigger product-line sync and remain zero after Save.
- Schedule must reside inside the actual Quote EditView form so data is submitted on Save; do not place it outside form tags or render a detached standalone template.

6. Campaign date arithmetic: must be exact

6.1 Definition

Weekly spots = sum of selected input quantities across all 168 fields for one theoretical week. Total spots = sum, for each selected field, of input quantity multiplied by the number of times that field's weekday occurs in the inclusive date range. Use timezone-safe calendar dates and reject an end date earlier than a start date.

```
weekly_spots = SUM(q[field] for every weekday/slot/hour field)
total_spots = 0
FOR every selected field:
    weekday = parse weekday from field name
    repeat_count = number of matching weekdays in [start, end]
    total_spots += quantity * repeat_count
```

Correctness and consistency are important: the same authoritative schedule-count algorithm must be used in the browser display, persisted Quote totals/product lines, conversion, and Myriad export. Prefer server-side authoritative calculation, with client-side preview using the identical business rules.

6.2 Required real-world examples

Date interval	Selected weekly cells	Expected
15 Jun 2026 - 24 Jun 2026	One on Tue, Wed, Thu, Fri (Morning Buzz)	Weekly spots = 4; Total spots = 6 (Tue 2, Wed 2, Thu 1, Fri 1)
18 Jun 2026 - 31 Jul 2026	Thu = 5 and Fri = 1	Weekly spots = 6; Total spots = 42 (7 Thursdays x 5 + 7 Fridays x 1)
Single Tuesday	Tuesday = 2; Wednesday = 3	Weekly spots = 5; Total spots = 2
Dates not provided	Any schedule values	Show weekly count; do not invent a campaign total or product quantity

> Known bug that was fixed
>
> NEVER calculate Total Spots = Weekly Spots x approximate number of weeks. Partial weeks, inclusive dates and exact weekday matches make that incorrect.

6.3 Implementation guidance: efficient weekday count

Use a deterministic routine: with start/end inclusive, compute the first occurrence of the target weekday on or after start, then if first <= end, count 1 + floor((end - first)/7 days). This avoids iterating a long range for every input. Verify Monday/Sunday mappings between JavaScript (getDay(): Sun=0) and PHP (date('w'): Sun=0).

```
function weekdayOccurrences(DateTimeImmutable $start,
                            DateTimeImmutable $end,
                            int $phpWeekday): int {
    $startW = (int) $start->format('w');  // Sunday = 0
    $offset = ($phpWeekday - $startW + 7) % 7;
    $first = $start->modify('+' . $offset . ' days');
    if ($first > $end) return 0;
    $days = (int) $first->diff($end)->format('%a');
    return 1 + intdiv($days, 7);
}
```

For zero-valued cells, skip calculation. Normalize dates to YYYY-MM-DD and use Africa/Johannesburg when converting input values to date-only objects. Never use browser UTC conversion that can shift selected calendar days.

7. Product catalogue and Quote pricing

7.1 Definitive products

Product name and part number	Unit price (ZAR)	Advert programme
PRE-BREAKFAST	300.00	05:00-06:00
MORNING-BUZZ	320.00	06:00-09:00
MIDMORNING-BANTER	350.00	09:00-12:00
MIDDAY-PULSE	350.00	12:00-15:00
CRUISE-CONTROL	350.00	15:00-18:00
SPECIALTY-SHOW-1	300.00	18:00-21:00
SPECIALTY-SHOW-2	220.00	21:00-00:00
MUSIC-UNLIMITED	120.00	00:00-05:00

Product names and part numbers have NO PUKFM prefix. Check for an existing non-deleted product matching part_number before inserting. The reported rates are the historical configuration, not a verified current tariff; make rates admin-configurable without breaking past Quote values.

7.2 One product line per programme

Selected schedule cells must be aggregated by programming slot/product. The Quote line quantity for that product is the total number of scheduled plays within the actual campaign date range, not merely that week's input count. Do not add one product line per hour or day unless the established UI specifically requires otherwise.

7.3 Price calculations and historical examples

Use SuiteCRM AOS_Quotes native calculation conventions and consistently handle VAT (historical South African 15% in the confirmed samples). Preserve whether the Quote form expects net line total or gross line total; do not blindly write VAT-inclusive values into a net field.

Recorded example	Calculation	Observed result
PRE-BREAKFAST, quantity 4	4 x R300 = R1,200; VAT R180	R1,380 including VAT
MORNING-BUZZ, quantity 2	2 x R320 = R640; VAT R96	R736 including VAT

Important Quote fields previously touched: aos_products_quotes.product_qty, product_unit_price, product_total_price, vat_amt, deleted. Quote EditView product element IDs included product_product_qty0, product_name0, product_part_number0, product_product_list_price0, product_product_unit_price0, product_vat_amt0, product_product_total_price0, product_item_description0, product_description0; totals include total_amt, subtotal_amount, tax_amount, total_amount. Inspect current markup because generated indexes may differ.

8. Automatic product synchronization: original design

8.1 Server-side logic hook

Original implementation file:

```
public/legacy/custom/modules/AOS_Quotes/RadioQuoteScheduleProducts.php
```

Hook class RadioQuoteScheduleProducts, method syncSelectedProducts. A previously inspected registry was:

```
public/legacy/custom/modules/AOS_Quotes/Ext/LogicHooks/logichooks.ext.php
```

Register the source hook through the correct custom/Extension/modules/AOS_Quotes/Ext/LogicHooks/ fragment, then run Quick Repair to regenerate the compiled registry. The earlier implementation used deferred/shutdown synchronization to handle Quote persistence; if rebuilding, carefully choose an after-save or deferred sync that can see Quote custom fields and avoids save recursion.

8.2 Required hook behaviour

1. Load actual Quote campaign dates and its 168 saved custom schedule fields; validate numeric nonnegative counts and dates.
2. Calculate quantity totals for each of the eight programmes across the inclusive campaign date range.
3. Locate corresponding AOS_Products by exact part number, with unambiguous handling for duplicates and missing catalog products.
4. For each programme with nonzero quantity, create or update one aos_products_quotes row with quantity, price, description, VAT and all native fields needed by SuiteCRM Quote totals.
5. For a programme reduced to zero, remove/soft-delete only its generated line item; preserve any unrelated manually entered Quote products.
6. Ensure repeated Save, Edit, conversion or AJAX sync does not duplicate generated lines. Use a stable way to identify schedule-created line items.
7. Recompute native Quote totals; ensure the final saved totals are identical to the displayed preview and PDF.

8.3 Idempotence and hook safety

- Do not create a new product row when the same product already has a schedule-generated row on this Quote. Update it instead.
- Do not recursively save the Quote from a hook without a guard. Confirm once-per-save behaviour and use transaction/locking where needed.
- For an existing Quote, changing campaign dates should recalculate all product quantities; changing a cell or using Clear All must reflect in DB and UI after Save.

9. Live AJAX scheduling integration

Previously existing entry-point source locations:

```
public/legacy/custom/Extension/application/Ext/EntryPointRegistry/
    radio_schedule_ajax_sync.php
public/legacy/custom/radio_schedule_ajax_sync.php
```

The weekly schedule template sent the selected inputs and date range to entryPoint=radio_schedule_ajax_sync. The endpoint returned the product-line data needed to update the visible Quote line editor and totals before Save.

9.1 Required AJAX contract

Request	Response / operation
Quote ID if editing	Verify access and identify saved record
Start/end dates	Normalize and validate inclusive range
Complete 168-field schedule payload	Calculate weekly and campaign totals
Optional response data	For each of eight programmes: part number, qty, price, VAT, net/gross line values, description
Client-side UI update	Map values to native Quote line rows and native summary/totals controls

All requests must be authenticated, CSRF-protected where applicable, ACL-checked, input-validated and safe from unintended DB changes during previews. If the native Quote UI uses different controls after SuiteCRM update, adapt to the actual DOM instead of hard-coding indexes.

9.2 Clear All and Save

Prior bespoke blocks were named RADIO_CLEAR_ALL_SPOTS_START/END and RADIO_SAVE_BUTTON_FIX_START/END. The latter resolved schedule Save problems. Reimplement their behaviour, not necessarily those exact source fragments.

- Clear All -> zero all 168 values -> recalculate -> update/remove generated product rows -> Save -> reload -> all zero.
- Do not intercept or replace Quote Save unless necessary. Maintain SuiteCRM validation and standard submit handlers.
- Debounce AJAX requests while typing and prevent stale responses from overwriting a newer schedule state.

10. Quote form placement, metadata and labels

10.1 Quote dates and labels

Add Quote DATE fields campaign_start_date_c and campaign_end_date_c to custom field storage and vardefs. Source extension paths previously used:

```
public/legacy/custom/Extension/modules/AOS_Quotes/Ext/Vardefs/
    pukfm_quote_schedule_dates.php
public/legacy/custom/Extension/modules/AOS_Quotes/Ext/Language/
    en_us.pukfm_quote_schedule_dates.php
```

User-facing labels: Campaign Start Date; Campaign End Date. Put them together near the schedule table and preserve entered values on validation errors and subsequent edits.

10.2 Embed the schedule in Quote EditView

Basic template path previously created:

```
public/legacy/custom/modules/AOS_Quotes/tpls/weekly_advert_schedule.tpl
```

Writing this file alone does NOT display it. Integrate it into the actual AOS_Quotes EditView metadata/customCode or recognized view-template inclusion path for SuiteCRM 8 legacy Quote forms. Inspect the generated DOM and prove all inputs belong to the Quote form. Do not duplicate unrelated Quotes fields or accidentally break the

native product line section.

10.3 Quote detail action

A working original Quote DetailView Action-menu entry titled Convert to Radio Campaign already existed. The stored customCode pointed to entryPoint=convert_quote_to_radio_campaign. Preserve this existing UI action instead of adding another visible button or a second conversion function.

```
public/legacy/custom/modules/AOS_Quotes/metadata/detailviewdefs.php
public/legacy/custom/Extension/application/Ext/EntryPointRegistry/
    convert_quote_to_radio_campaign.php
public/legacy/custom/convert_quote_to_radio_campaign.php
```

11. Clean Quote PDF output

A key user request was to keep the schedule data in CRM but NOT print verbose grid summaries into the Quote PDF or product descriptions. Previously unwanted strings included Schedule Summary and Selected Weekly Schedule.

11.1 Product description format

Approved example:

```
One 30 second prerecorded advertisement played between 15:00 and 18:00.
```

The product-line description should contain a natural-language description for that programme only. Remove any auto-generated week/day list or schedule summary block before persistence and PDF rendering. Previously named cleanup function: cleanRadioLineDescription in RadioQuoteScheduleProducts.php.

11.2 PDF test

- Create a Quote with two or more programmes, generate/download the native AOS_Quotes PDF, and check every product description.
- Assert “Schedule Summary” and “Selected Weekly Schedule” do not appear anywhere in the generated PDF or persisted schedule-generated description columns.
- Do not remove native Quote totals, customer details, VAT information or required quotation headers.

12. Convert Quote to existing Radio Campaign

12.1 Action semantics

The Quote DetailView must offer one action Convert to Radio Campaign. Its entry point reads the Quote record identifier, loads AOS_Quotes, and either finds the already-linked campaign or creates one new campaign. Repeated conversion of the same Quote must open/update the same campaign, not create duplicates.

12.2 Lookup and field mapping

Source / logic	Target field
"Radio Campaign - " + Quote name	campaign.name
campaign_start_date_c	campaign.start_date
campaign_end_date_c	campaign.end_date
Exact schedule total over dates	campaign.number_of_spots
Quote number/reference	campaign.quote_reference
Quote id	campaign.linked_quote_id
Quote assigned_user_id	campaign.assigned_user_id
Default if new/empty	campaign.spot_length = 30 sec
Default if new/empty	campaign.status = Draft

Original working lookup attempted linked_quote_id, custom-field variant linked_quote_id_c when present, and fallbacks such as Quote reference. Reconstruct defensively, giving the exact Quote-ID link precedence and avoiding ambiguous matches.

12.3 Conversion test evidence

In the original working environment a test Quote named Test Radio Advert Quote converted to Radio Campaign - Test Radio Advert Quote, with start 2026-06-15, end 2026-06-27, number_of_spots = 6, spot_length = 30 sec, status = Draft and quote_reference = 5. Myriad cart was initially NULL then set to 1 for export.

12.4 Additional safeguards

- Validate Quote ACLs, start/end dates and schedule totals before conversion. Redirect safely to the resulting campaign.
- Preserve an already-entered Myriad cart number, manually edited status or spot duration unless the business explicitly requests an overwrite.
- Relationship join-table record must not be duplicated on every conversion.

13. SuiteCRM 8 Export to Myriad record action

13.1 Why the UI placement matters

The original export initially suffered from an absent or blank UI control when implemented as a legacy DetailView button. The working implementation used a SuiteCRM 8 recordActions configuration and Symfony process service under the defaultExt extension. A legacy templateMeta form button is not an equivalent substitute.

13.2 Config/service locations

```
extensions/defaultExt/config/modules/RT_RadioCampaigns/
    recordview/actions/export-to-myriad.php
extensions/defaultExt/modules/RT_RadioCampaigns/Process/Service/
    RecordActions/ExportToMyriadAction.php
```

13.3 Known handler signature

Class namespace: App\Extension\defaultExt\modules\RT_RadioCampaigns\Process\Service\RecordActions. Class name: ExportToMyriadAction. It extended LegacyHandler, implemented ProcessHandlerInterface, and used the process type record-radio-campaign-export-to-myriad. Methods included getProcessType(), getHandlerKey(), requiredAuthRole(), getRequiredACLs(), configure(), validate(), run().

The record action key was radio-campaign-export-to-myriad, with LBL_EXPORT_TO_MYRIAD, detail mode, confirmation UI and an authenticated process. The exact config loader array shape must be verified against SuiteCRM 8.10.1 in this deployment.

> Critical fresh-script defect
>
> The fresh-install script placed a require of public/legacy/custom/export_radio_campaign_to_myriad.php inside the Symfony handler. The legacy script calls SugarApplication::redirect or can terminate execution. This is NOT equivalent to the previously working full service and may break the asynchronous response. Put export logic in a shared library/service and return a proper process result; do not depend on a redirecting legacy entry point.

13.4 Required service workflow

1. Initialize SuiteCRM legacy context using the supported handler pattern; authenticate and authorize view/export access for the campaign.
2. Resolve the campaign record and require a valid Myriad cart number and linked Quote.
3. Read date range and the Quote weekly grid, expand actual scheduled dates, compute exact scheduled times, and generate the CSV/daily log files.
4. Write files atomically and validate number of records; only mark status exported_to_myriad after successful writing.
5. Close legacy context and return success/error through the framework process interface; allow page refresh and show a useful notification.

14. Myriad export specification

14.1 Preconditions and storage

Exporting requires: a Radio Campaign with start/end dates, a linked source Quote with nonzero selected scheduled spots, and myriad_cart_number set. Write outputs to:

```
/srv/myriad_exports/
/srv/myriad_exports/myriad_daily_logs/
```

The export previously produced one detailed CSV covering every play and one daily .log file for each active day with selected plays. Filenames include campaign and timestamp to avoid overwriting prior exports.

14.2 Detailed CSV header - exact sequence

```
Campaign Name,Quote Reference,Myriad Cart Number,Date,Day,
Scheduled Time,Start Time,End Time,Slot,Spot Length,
Play Number,Total Plays In Block
```

One CSV row per actual airing: a selected Thursday field with value 2 generates two rows for EACH Thursday in the campaign date range, not a row for a weekly allocation.

14.3 Example CSV rows (original verified pattern)

```
"Radio Campaign - Test Radio Advert Quote",5,1,2026-06-16,
Tuesday,21:30:00,21:00,22:00,"Specialty Show 2","30 sec",1,1
"Radio Campaign - Test Radio Advert Quote",5,1,2026-06-18,
Thursday,04:20:00,04:00,05:00,"Music Unlimited","30 sec",1,2
```

14.4 Time spreading within a one-hour block

Historical implementation used evenly spaced positions interval = floor((end_seconds - start_seconds)/(N + 1)); play j at start_seconds + interval * j. An hour containing two plays therefore schedules 04:20:00 and 04:40:00. A single play in an hour schedules at :30:00. The 23:00-00:00 block crosses midnight correctly for duration math.

Scheduling is a deterministic planning choice, not proof that an external Myriad log importer accepts every time. Verify Myriad-side specifications and collision behaviour before production delivery.

14.5 Daily Myriad .log record format

```
| |HH:MM:SS|CART|MM:SS|AUDIO|TITLE|
```

Example for a cart number 1, 30-second advert, Music Unlimited block:

```
| |04:20:00|1|00:30|AUDIO|Radio Campaign - Test Radio Advert Quote - Music Unlimited|
| |04:40:00|1|00:30|AUDIO|Radio Campaign - Test Radio Advert Quote - Music Unlimited|
```

File naming pattern:

```
PUKfm_Adverts_YYYYMMDD_{safe campaign name}_{timestamp}.log
```

Make filename-safe campaign names; remove pipes and CR/LF from the log title. Use CRLF line endings if required by the specific Myriad import; the original implementation wrote CRLF. Preserve the ordering and exact delimiter count.

14.6 Previously observed export example

In a reported successful test, the CSV was created under /srv/myriad_exports/radio_campaign_Radio_Campaign_-_Test_Radio_Advert_Quote_20260617_183041.csv. Daily logs included 2026-06-16, 2026-06-18, 2026-06-23 and 2026-06-25. Do not treat those historical files as guaranteed present now.

14.7 Myriad compatibility is a separate gate

The original test demonstrated local CSV and .log file creation, not final acceptance inside every Myriad configuration. Before production, import one daily file into the target Myriad system and verify the expected cart, play type, timing and traffic format. Adjust the file dialect only after evidence from the target importer.

15. Optional Windows/Samba export access

A previously discussed share made /srv/myriad_exports accessible as \\92.4.131.128\myriad_exports. This is an optional convenience, not a prerequisite for the CSV/log generation. Never expose guest-writable Samba (port 445) to the public internet.

15.1 Recommended secure deployment

- Prefer authenticated Samba reachable only over an internal network, VPN or locked-down source IPs, or transfer via SFTP.
- Map an approved user/group to /srv/myriad_exports and ensure www-data can write while authorized operators can read.
- Use one test daily export first; confirm whether Myriad requires a local path rather than UNC share.

16. Database initialization and safe migration design

16.1 Expected tables

Table	Expected reason
aos_quotes	SuiteCRM standard Quote records (must already exist)
aos_quotes_cstm	Quote campaign dates and 168 schedule values
aos_products	Eight schedule product catalogue records
aos_products_quotes	Product rows associated with saved Quotes
rt_radiocampaigns	Radio Campaign records
rt_radiocampaigns_cstm	Optional custom field extension
rt_radiocampaigns_aos_quotes_c	Campaign-to-Quote relationship records

16.2 Migration principles

- Inspect information_schema for each table and column before changing it; make migrations idempotent.
- The previous script first failed because aos_quotes_cstm did not exist on the clean database. Ensure it exists with a primary key id_c when required, but respect the installed SuiteCRM custom-field mechanism.
- The old script used ALTER TABLE ... ADD COLUMN IF NOT EXISTS and succeeded on the reported MariaDB build; prefer capability checks and explicit column probes for cross-version safety.
- Back up the database before changes; never DROP or reset an installed production database as part of Radio CRM customization.

16.3 Verify data columns and counts

```
sudo mysql suitecrm_db -e "SHOW TABLES LIKE 'aos_quotes_cstm';"
sudo mysql suitecrm_db -e "SHOW TABLES LIKE 'rt_radiocampaigns';"
sudo mysql suitecrm_db -e "SHOW COLUMNS FROM aos_quotes_cstm;"
sudo mysql suitecrm_db -e "SELECT COUNT(*) FROM aos_products
  WHERE part_number IN ('PRE-BREAKFAST','MORNING-BUZZ',
  'MIDMORNING-BANTER','MIDDAY-PULSE','CRUISE-CONTROL',
  'SPECIALTY-SHOW-1','SPECIALTY-SHOW-2','MUSIC-UNLIMITED')
  AND deleted=0;"
```

Require eight distinct catalog part numbers, exactly 168 schedule columns, the two campaign DATE columns, a valid Radio Campaign bean/table and a functioning Quote relationship.

17. Previous restore failures and how NOT to repeat them

Symptom	Cause	Required correction
Download succeeded but "Could not find extracted SuiteCRM folder"	ZIP was extracted flat, not inside SuiteCRM-versioned folder	Discover installation root by bin/console or composer.json, then copy safely
PuTTY lost/garbled pasted script lines	Huge heredoc pasted through terminal poorly	Upload a version-controlled archive/script via SCP/SFTP or fetch a trusted artifact; use short commands
ERROR 1146: aos_quotes_cstm does not exist	Fresh SuiteCRM had no Quote custom table yet	Create via repair/migration then apply custom fields
Could not write custom/metadata/*.php	Parent directory missing	mkdir -p all required directories before writing files
Could not write Ext/Language/*.php	Another parent directory missing	Pre-create full directory tree; fail preflight before modifying files
sudo restore script.sh. command not found	Trailing dot after .sh	Run exact pathname without punctuation
Script paused after PHP syntax checks	Permissions/cache clear can be slow; user interrupted with Ctrl+C	Watch processes/logs; validate completion explicitly; do not trust an echo after interrupt
SuiteCRM worked locally but public webpage timed out	Likely firewall, route, instance public IP or Apache binding	Separate local HTTP test from cloud ingress/network connectivity
Export action invisible/blank	Wrong legacy DetailView button placement	Use SuiteCRM 8 defaultExt recordview action + process service
Fresh script appeared to complete	It omitted product sync and did not embed the schedule template	Run UI/end-to-end tests; rebuild missing functionality

Avoid the giant recovery script as production source of truth

An example script at /usr/local/sbin/restore_radio_crm_custom.sh was saved on the fresh install, but it is only a partial reconstruction. It is not an old backup, and it may have incomplete UI wiring, wrong Symfony service behaviour and non-idempotent conversion. Audit it before reuse.

18. Backup, rollback and deployment strategy

18.1 Historical backup workflow

A previous backup script was planned/created at /usr/local/sbin/create_radio_crm_backup.sh, storing snapshots under /srv/radio_crm_backups/radio_crm_backup_TIMESTAMP with a portable tar.gz and a symbolic link latest_radio_crm_backup. Files included custom legacy directories, RT_RadioCampaigns module code, defaultExt configs/process handlers and Myriad exports; database was dumped to a compressed SQL file.

18.2 Historical restore commands (only if backup actually exists)

```
cd /srv/radio_crm_backups/latest_radio_crm_backup
sudo ./restore_radio_crm.sh                # custom files only
sudo ./restore_radio_crm.sh --with-db      # files + DB, HIGH RISK
```

The above paths/commands are historical. On the reported new server, those files are NOT proven to exist. Before using --with-db, inspect schema/version and database content, obtain approval, and create a fresh backup. Never overwrite new CRM data without an explicit decision.

18.3 Recommended agent delivery

- Build an install_radio_crm.sh plus numbered SQL/PHP migrations and real application files in Git; avoid embedding hundreds of file definitions in one heredoc.
- Provide preflight.sh, backup.sh, deploy.sh, smoke_test.sh and rollback documentation.
- Keep changes staged and recorded; check git diff before file overwrites; backup each modified base file and use supported extension overrides.
- Publish a manifest of installed/changed files with SHA-256 digests and include a final acceptance report.

19. Build order for the local AI agent

Each phase is a checkpoint. Do not advance until the tests listed for that phase pass.

Phase	Work package	Gate
0 - Discovery	Inventory installation, original backups, current code, module schema, UI framework	Signed-off inventory and backup
1 - Module	Repair/register RT_RadioCampaigns, vardefs, language, menus, relationship	Menu and List/Edit/Detail all work
2 - Quote fields	Create campaign dates and 168 fields; add labels	Persistence and re-open test pass
3 - Schedule UI	Place grid in Quote EditView; summaries; Clear All	All cells inside form, totals correct
4 - Products	Catalogue, hook, AJAX sync, native Quote totals	Prices/VAT/rows correct and idempotent
5 - PDF	Clean product descriptions, normal Quote PDF	No verbose schedule text
6 - Convert	Existing action, lookup, link, dates and count	Repeated conversion does not duplicate
7 - Export	SuiteCRM 8 record action, handler, CSV and .log files	Correct counts/times/logs; success notification
8 - Release	Backups, security, cache repair, regression suite	Documented reproducible build

20. End-to-end acceptance tests (mandatory)

20.1 Quote edit and schedule calculation

A1. Create Quote with 15 Jun to 24 Jun 2026; set one Tue/Wed/Thu/Fri Morning Buzz input (any one hour within slot each); assert Weekly = 4 and Total = 6.

A2. Save and reopen; the four selected values, both dates and totals remain unchanged.

A3. Change dates to 18 Jun to 31 Jul 2026; set Thu=5 and Fri=1 only; assert Weekly = 6, Total = 42.

A4. Zero all schedule cells via Clear All; Save and reload; no generated schedule product lines remain, and all values persist as zero.

A5. Reject invalid/negative/noninteger quantities; display sensible validation for end date before start date.

20.2 Quote products and PDF

B1. Select Pre Breakfast 4 total plays and Morning Buzz 2 total plays; assert two correct product lines, rates R300/R320, net R1,200/R640 and VAT R180/R96 if VAT=15%.

B2. Save twice; assert no extra generated line rows. Add a manual non-radio product; assert it survives schedule edits.

B3. Generate Quote PDF; line descriptions read as natural-language adverts and omit Schedule Summary / Selected Weekly Schedule.

20.3 Radio Campaign conversion

C1. Convert a saved Quote; assert name, dates, total spots, reference and linked quote ID.

C2. Convert SAME Quote again; assert same campaign ID, no duplicate relationship, existing cart number and user-entered settings preserved.

C3. Edit schedule/dates on Quote, reconvert; assert campaign accurately reflects changes without losing unrelated details.

20.4 Export correctness

D1. Set cart to 1 and duration to 30 sec. Export campaign from the SuiteCRM 8 Actions menu; assert user-visible success message.

D2. Check the CSV header matches the 12 columns and there is one data row per actual play.

D3. Two plays within 04:00-05:00 must generate 04:20:00 and 04:40:00 on each selected applicable date.

D4. Each applicable date produces a separate .log, correctly formatted, with cart and 00:30 duration.

D5. Missing cart, missing Quote, zero spots, invalid dates and failed file writes must produce errors without setting status to exported_to_myriad.

D6. Repeat export; result should be a separate timestamped snapshot without destroying prior files.

20.5 Upgrade and permissions

E1. Quick Repair and Rebuild does not erase custom actions, hooks, fields, or schedule template.

E2. Syntax checks pass, logs show no unexpected PHP errors, and the Apache worker can write export files.

E3. A non-admin user with only approved CRM permissions can access the appropriate features, but cannot bypass ACL to export others' restricted records.

21. Diagnostics and validation commands

21.1 Health and repair

```
cd /var/www/html
sudo apachectl configtest
sudo systemctl status apache2 mariadb --no-pager
curl -I http://127.0.0.1
sudo -u www-data php bin/console cache:clear
sudo journalctl -u apache2 -n 80 --no-pager
sudo tail -n 100 /var/log/apache2/suitecrm-error.log
```

From admin UI use Admin -> Repair -> Quick Repair and Rebuild. Review and execute generated SQL where needed. Prefer scheduled quiet time for production upgrades.

21.2 Locate custom logic

```
cd /var/www/html
sudo grep -Rin 'convert_quote_to_radio_campaign'   public/legacy/custom | head -40
sudo grep -Rin 'radio_schedule_ajax_sync'   public/legacy/custom | head -40
sudo find extensions/defaultExt -type f | grep -E 'Myriad|myriad'
sudo php -l public/legacy/custom/convert_quote_to_radio_campaign.php
sudo php -l public/legacy/custom/export_radio_campaign_to_myriad.php
```

21.3 Database checks

```
sudo mysql suitecrm_db -e "SELECT id,name,start_date,end_date,number_of_spots,
 spot_length,status,myriad_cart_number,quote_reference,
 linked_quote_id,deleted
 FROM rt_radiocampaigns WHERE deleted=0
 ORDER BY date_modified DESC LIMIT 5;"
```

21.4 Export files

```
sudo find /srv/myriad_exports -maxdepth 1 -type f   -name '*.csv' -printf '%TY-%Tm-%Td %TH:%TM %p\n' | sort
sudo find /srv/myriad_exports/myriad_daily_logs   -maxdepth 1 -type f -printf '%p\n' | sort
sudo ls -ld /srv/myriad_exports /srv/myriad_exports/myriad_daily_logs
```

Sensitive credentials must not be pasted into logs or agent responses. For production error reports, provide sanitized SQL results and PHP stack traces with secrets redacted.

22. Known unknowns the agent must resolve

- The exact original Quote EditView schedule injection mechanism and original working template JavaScript were not recovered as a complete source archive.
- The historical working full ExportToMyriadAction.php body is described by behaviour and test output; the later simplified new-server script is not equivalent.
- The exact product-line field semantics (net/gross columns) need inspection of the installed AOS_Quotes logic, despite known arithmetic examples.
- Whether the backup tar.gz and old database still exist on the new server is unknown; never label the partial restore script as a backup.
- External access and old/myriad network shares depend on the current OCI public IP, VCN ingress and local firewall; do not assume the last-supplied IP is permanent.

- The final Myriad importer dialect and whether it accepts the generated .log format in the target instance must be tested on the actual playout system.
- Custom module ACL roles, portal exposure and long-term quote status transitions were not fully specified; request confirmation before broadening access or changing workflows.

23. Master prompt for a locally hosted AI engineering agent

> How to use this section
>
> Supply the entire PDF to the agent as reference. Give the following master prompt as its top-level task. The agent should inspect the live system rather than inventing files or declaring success from terminal output alone.

ROLE AND OBJECTIVE

You are the senior SuiteCRM 8 developer and systems engineer responsible for reconstructing the PUKfm Radio CRM implementation described in this PDF. Work against the actual Ubuntu server. Your deliverable is a running, tested SuiteCRM 8 custom radio advertising workflow, plus source-controlled installation/upgrade scripts and a concise verification report. This document is a behavioural specification, NOT a guarantee that legacy source files still exist.

MANDATORY OPERATING RULES

1. Start by gathering version, filesystem, database, Apache and networking facts. Search for real backups and working code. Report existing vs missing components.
2. Create a complete application/database backup and rollback plan before any schema or source modification. Do not drop data. Require approval for destructive changes.
3. Inspect native SuiteCRM 8.10.1 metadata, form rendering, record actions, auth and process interfaces. Implement with supported extension points and no duplicate buttons.
4. Build the 168 weekly grid fields and two dates, implement inclusive weekday occurrence calculations, live preview, Save persistence, Clear All, dynamic eight-product line sync and clean PDF descriptions.
5. Restore the existing Quote-to-Radio-Campaign action and make it idempotent; ensure proper linking and accurate totals.
6. Implement a real SuiteCRM 8 Export to Myriad action and backend service that generates scheduled-play CSV + daily .log files. Do not call redirecting legacy entry points inside an async process.
7. Use safe migrations, explicit validation, transactions where appropriate, secure permissions, tests and source-control commits. Capture command output and material errors.
8. At each phase demonstrate successful UI behaviour, matching database state, and—where applicable—file output. Failure must block completion of that phase.
9. Do not paste large unverified one-shot scripts into PuTTY; transfer implementation files via SCP/SFTP/Git, then run concise versioned installation commands.
10. When finished, provide a file-change manifest, installed module details, database migration log, tested scenarios with observed results, security notes and rollback instructions.

ACCEPTANCE STANDARD

The implementation is accepted only when all tests in Section 20 pass, the original eight price points and 24-hour slot map are preserved, no extra campaign records or Quote rows appear after repeat actions, and the Myriad export files match the stated format and row counts.

24. Final operator checklist

- ☐ Base SuiteCRM 8 installed, reachable, authenticated and backed up.
- ☐ Radio Campaigns menu and all three screens (List/Edit/Detail) function.
- ☐ Exactly 168 schedule fields plus 2 dates exist and persist in Quote records.
- ☐ Schedule is embedded into the existing Quote EditView within the form.
- ☐ Weekly and campaign totals reflect exact inclusive weekdays.
- ☐ Eight product records have correct part numbers/rates and sync automatically.
- ☐ Clear All, Edit/Save and repeat Save are safe and idempotent.
- ☐ Quote PDF has clean product descriptions and correct native totals.
- ☐ Existing Convert action updates, rather than duplicates, linked Campaigns.
- ☐ Export to Myriad displays as a proper SuiteCRM 8 Actions-menu entry.
- ☐ CSV and daily .log files have correct play counts, times, names and formatting.
- ☐ No export success/status update occurs on validation or write failures.
- ☐ ACLs, filesystem permissions, cloud ingress and secrets are checked.
- ☐ Backup, rollback package, source manifest and tested installer are delivered.

End of handover. This document preserves the known configuration and user-required behaviour. It explicitly identifies gaps that cannot truthfully be claimed as recovered source code.